# AQ3.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.

**Level 1**

*1 point*

Choose the set of correct options.

- [ ] The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.
- [ ] The set $\{(1,2,3),(4,5,6),(5,7,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.
- [ ] The set $\{(1,2,3),(0,0,6),(0,0,9)\}$ is linearly independent.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The set $\{(1,2,3),(4,5,6),(7,8,9)\}$ is linearly dependent.

The set $\{(1,2,3),(0,5,6),(0,0,9)\}$ is linearly independent.

*1 point*

If $ad-bc$ = 0ad − bc = 0 then which of the following options are true?

- [ ] The set $\{(a,b),(c,d)\}$ is linearly dependent in $R^2$ R2.
- [ ] The set $\{(a,c),(b,d)\}$ is linearly dependent in $R^2$ R2.
- [ ] The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in $R^3$ R3.
- [ ] The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in $R^3$ R3.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The set $\{(a,b),(c,d)\}$ is linearly dependent in $R^2$ R2.

The set $\{(a,c),(b,d)\}$ is linearly dependent in $R^2$ R2.

The set $\{(1,0,0),(0,a,b),(0,c,d)\}$ is linearly dependent in $R^3$ R3.

The set $\{(1,0,0),(0,a,c),(0,b,d)\}$ is linearly dependent in $R^3$ R3.

For $c \in Z$c ∈ Z, consider the set of three vectors $S$ = $\{(1,c,0),(c,0,c),(2c,3c,4c)\}$S = {(1, c, 0), (c, 0, c), (2c, 3c, 4c)} in $R^3$ R3 with usual addition and scalar multiplication. Find the value of $c$c such that the given set is linearly dependent.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

*1 point*

*1 point*

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$\{(a1, b1, c1), (a2, b2, c2), (a3, b3, c3)\} is given to be linearly independent and consider the following system of linear equations

$$a_1 x + b_1 y + c_1 z = 1$$
$$a_2 x + b_2 y + c_2 z = 2$$ a1x + b1y + c1z = 1a2x + b2y + c2z = 2a3x + b2y + c3
$$a_3 x + b_2 y + c_3 z = 3.$$

z = 3.

If $AX = b$AX = b is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$X = $\begin{bmatrix} \\ \\ \end{bmatrix}$xyz, then which of the following option(s) is(are) true?

- ☐ The system of linear equations has a unique solution.
- ☐ The system of linear equations has no solution.
- ☐ The system of linear equations have infinitely many solutions.
- ☐ The coefficient matrix $A$A is invertible.
- ☐ The coefficient matrix $A$A is not invertible.
- ☐ Any one of the above options cannot be concluded using the given information.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The system of linear equations has a unique solution.
The coefficient matrix $A$A is invertible.

*1 point*

Assume the set $\{(a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3)\}$\{(a1, b1, c1), (a2, b2, c2), (a3, b3, c3)\} is given to be linearly dependent and consider the following system of linear equations

$$a_1 x + b_1 y + c_1 z = 1$$
$$a_2 x + b_2 y + c_2 z = 2$$ a1x + b1y + c1z = 1a2x + b2y + c2z = 2a3x + b2y + c3
$$a_3 x + b_2 y + c_3 z = 3.$$

z = 3.

If $AX = b$AX = b is matrix representation of the above system of linear equations, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$X = $\begin{bmatrix} \\ \\ \end{bmatrix}$xyz, then which of the following option(s) is(are) true?

- ☐ Either the system of linear equations has a unique solution or no solution.
- ☐ Either the system of linear equations has no solution or infinitely many solutions.
- ☐ Either the system of linear equations have infinitely many solutions or a unique solution.
- ☐ The system of linear equations has a unique solution.
- ☐ The coefficient matrix $A$A is invertible.
- ☐ The coefficient matrix $A$A is not invertible.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**
Either the system of linear equations has no solution or infinitely many solutions.
The coefficient matrix $A$A is not invertible.
If $S$S is a linearly independent set in vector space $R^3$R3 with respect to usual addition and scalar multiplication, then find the maximum possible cardinality of $S$S.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 3
*1 point*

**Level 2**

*1 point*
Let the set $\{(a,b),(c,d)\}${(a, b), (c, d)} be a linearly independent subset of $R^2$R2. Choose the set of correct options.

☐ $\{(a,b,0),(c,d,0)\}${(a, b, 0), (c, d, 0)} must be a linearly independent subset of $R^3$R3.

☐ $\{(a,b,0),(c,d,0),(0,0,1)\}${(a, b, 0), (c, d, 0), (0, 0, 1)} must be a linearly independent subset of $R^3$R3.

☐ $\{(a,b,0),(c,d,0),(1,0,0)\}${(a, b, 0), (c, d, 0), (1, 0, 0)} must be a linearly independent subset of $R^3$R3.

☐ $\{(a,b,0),(c,d,0),(0,1,0)\}${(a, b, 0), (c, d, 0), (0, 1, 0)} must be a linearly independent subset of $R^3$R3.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(a,b,0),(c,d,0)\}${(a, b, 0), (c, d, 0)} must be a linearly independent subset of $R^3$R3.
$\{(a,b,0),(c,d,0),(0,0,1)\}${(a, b, 0), (c, d, 0), (0, 0, 1)} must be a linearly independent subset of $R^3$R3.
*1 point*
Each column of an $n \times n$n × n matrix $A$A can be thought of as vectors in $R^n$Rn. If a set $S$S consists of all the column vectors of $A$A, then which of the following is(are) true?

☐ $S$S is always linearly independent.

☐ $S$S is linearly independent if $det(A) \neq 0$det(A) ≠ 0.

☐ $S$S is linearly independent if $det(A) = 0$det(A) = 0.

☐ If $det(A) = 1$det(A) = 1, then $S$S is linearly independent.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$S$S is linearly independent if $det(A) \neq 0$det(A) ≠ 0.
If $det(A) = 1$det(A) = 1, then $S$S is linearly independent.
*1 point*

Suppose it is known that the three vectors $v_1 = [a, b, c], v_2 = [d, e, f], v_3 = [g, h, i]$v1 = [abc], v2 = [def], v3 = [ghi] are linearly dependent in $R^3$ R3. Which of the following is correct?

- [ ] Some linear combination of $a, b, c$ a, b, c must be 0.
- [ ] Some linear combination of $a, d, g$ a, d, g must be 0.
- [ ] Some linear combination of $d, e, f$ d, e, f must be 0.
- [ ] Some linear combination of $c, f, i$ c, f, i must be 0.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
Some linear combination of $a, b, c$ a, b, c must be 0.
Some linear combination of $a, d, g$ a, d, g must be 0.
Some linear combination of $d, e, f$ d, e, f must be 0.
Some linear combination of $c, f, i$ c, f, i must be 0.

If vectors $(2x, 4, -6), (-6, -3x, 2x), (-1, x-3, x+2)$(2x, 4, −6), (−6, −3x, 2x), (−1, x − 3, x + 2) from the vector space $R^3$ R3 with usual addition and scalar multiplication, are linearly dependent and $x \in [2, 5]$x ∈ [2, 5], then find the value of $x$x.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
(Type: Numeric) 2
*1 point*